IILM UNIVERSITY, GREATER NOIDA

# NetSage AI

*AI-Assisted Network Fault Diagnosis with Enforced Human Review*

Cisco – AICTE Virtual Internship Program 2026 • Problem Statement 2: Applied AI + Network Troubleshooting


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PROGRAMME & SEMESTER
**B.Tech CSE • Semester 5**

SESSION
**2024 – 2028**

ORGANIZATION
**Cisco Networking Academy (AICTE VIP 2026)**

DOMAIN
**Applied AI • Modern AI track**

## 01 The Problem & Objective

**Diagnosis is slow and manual**
An engineer reads pages of show-command output and infers the fault by hand.

**LLMs can help — but they hallucinate**
They invent evidence or reach a confident, wrong conclusion.

**So an unsupervised AI troubleshooter is unsafe**
It cannot be trusted to act on a live network by itself.

OBJECTIVE

**Use an LLM's reasoning to accelerate fault diagnosis — while guaranteeing that a human validates every fix before it can be applied.**

*The reasoning of AI, with the accountability of human review.*

# 02 What NetSage AI Does

**INPUT**

A plain-language fault report + the show-command output from the affected devices.

**NETSAGE AI**

Analyzes the transcript, cross-checks two independent layers, and explains the mechanism.

**STRUCTURED DIAGNOSIS**

Root cause • OSI layer • verbatim evidence • next command • fix steps.

## Then it STOPS.

No fix it proposes can be applied until a human reviewer approves it — and that gate is enforced in code, not written in a policy document. There is no override flag.

## 03 How It Works — Four Stages

### 1 Rule Checker

34 deterministic checks in pure Python. Every finding quotes the exact transcript line — it cannot hallucinate.

### 2 AI Diagnosis

Claude reasons across multiple show outputs. Output is schema-validated and every evidence line is checked verbatim.

### 3 Human Review

Presents transcript → checker → AI, in that order. The reviewer accepts, edits, or rejects.

### 4 Apply-Gate + Dashboard

The apply step refuses fixes unless approved. A pandas / matplotlib dashboard summarises coverage.

*The two automated layers are designed to fail in opposite directions.*

04

# Two Layers, Failing in Opposite Directions

## RULE CHECKER

✓ Cannot hallucinate — every finding carries its verbatim trigger line.

✗ Cannot reason about intent — it misses NS-002, a port in the wrong VLAN where every config line is well-formed.

VS

## AI DIAGNOSIS

✓ Can reason across many show outputs and explain the mechanism in prose a student can learn from.

✗ Can invent — on NS-021 it imagined powered-off hosts to explain a reachability boundary at .63.

*Where they disagree, one of them is wrong — a far more useful thing to hand a reviewer than a single confident answer.*

# 05 Human Oversight — Enforced in Code, Not Policy

```
$ python src/netsage.py apply --case NS-031

!! REFUSED — no fix steps will be emitted for NS-031.

NS-031 was REJECTED on review by Samarth Mehrotra.
The AI diagnosis was found incorrect, so its fix
steps must not be applied.

Reviewer note: 'State Assoc' means the WPA2 four-way
handshake already completed — so the link-layer fix
cannot be right.
```

- Evidence is verified against the transcript before a human sees anything.
- The reviewer reads the raw evidence before the AI's conclusion — by design.
- No override flag. No code path from a diagnosis to a fix that skips the gate.

06

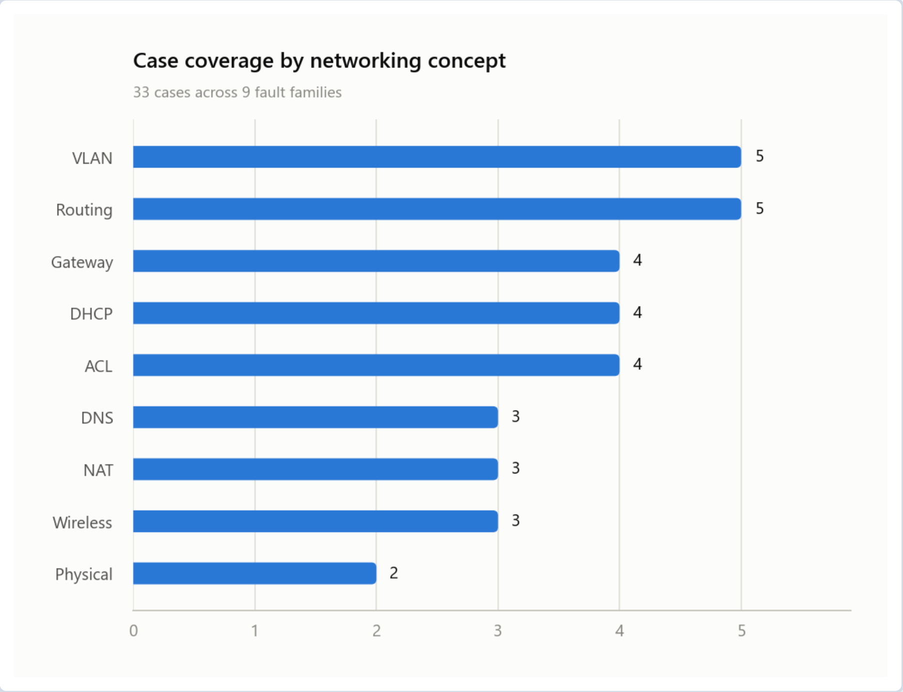
# Results — Reproducible from the CLI

**33**
Troubleshooting cases · 9 fault families

**34**
Deterministic checks · 97% catch rate

**33 / 33**
AI diagnoses schema-valid

**65**
Automated tests passing




## Coverage & honesty

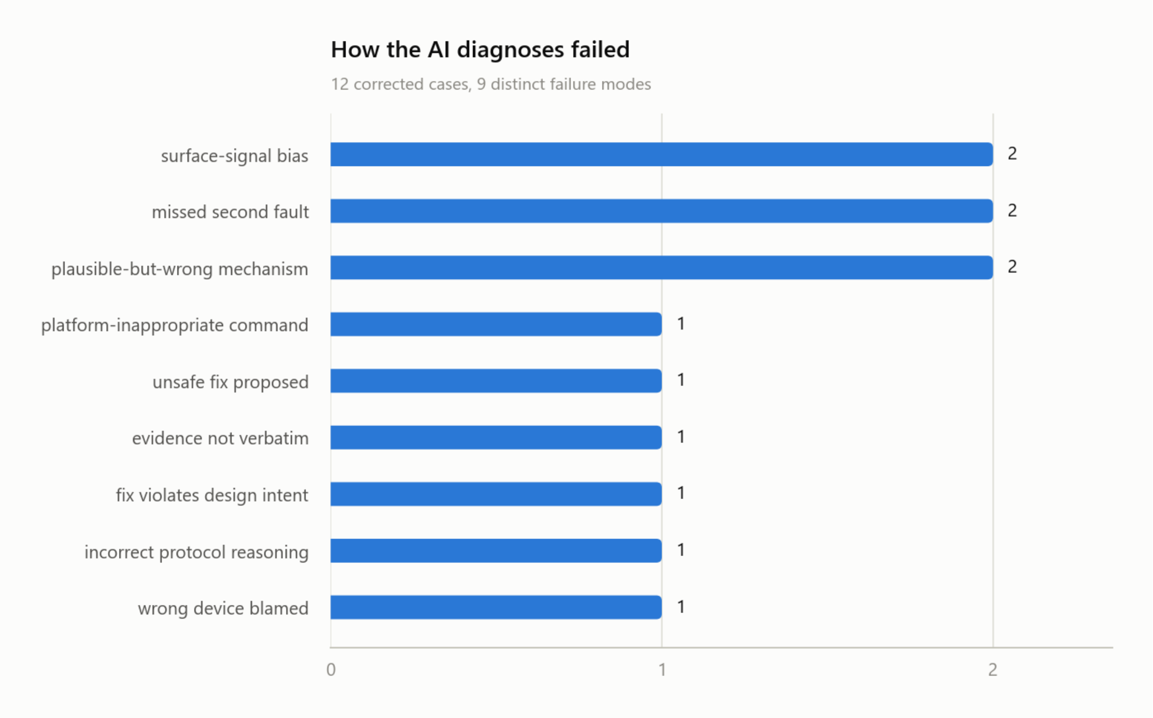
- 9 fault families across OSI layers 1–4 and 7.

Brief required 30 cases — 33 delivered.

Brief required 5 corrections — 12 documented.

Every figure is produced by code in the submitted project and reproducible from one command.

# 07 The Result Worth Being Honest About




**63.6%**

AI–human agreement (21 accepted · 7 edited · 5 rejected)

The 12 non-accepted diagnoses are committed as-is, each documented with its failure mode — not regenerated until the model agreed.

### Reported honestly

This is a demonstration corpus, not a blind benchmark — the case library and the recorded AI responses were produced in the same session. A --provider anthropic path exists for a genuinely blind re-run on a fresh case set.

# 08 Responsible AI in Practice

## Human oversight here is mechanical, not advisory.

### 1 Verify first

Evidence is checked against the transcript before any human sees it.

### 2 Evidence before answer

The reviewer reads the raw evidence before the AI’s conclusion — by design.

### 3 The gate blocks bad fixes

Unreviewed or rejected diagnoses can never emit fix steps.

**12**

corrections logged — each with its failure-mode category and the prompt, checker, or procedure change it drove. Most were not fixable by prompting the model more nicely, which is itself a finding.

# 09 Technology & Testing

### Python 3.13

Pure Python — no heavy framework, no cloud stack to stand up.

### pandas + matplotlib

Dashboard charts, with a palette validated for colour-blind safety.

### Jinja2

A single, self-contained HTML dashboard report.

### Claude API

Called directly over HTTPS via requests — no SDK dependency.

### Offline by default

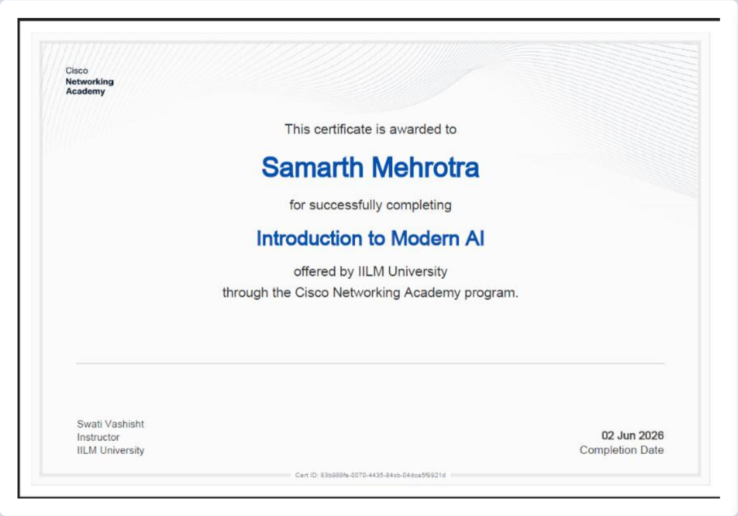
A recorded provider replays committed responses — a demo can't fail on connectivity.

### 65 pytest tests

Including named regression tests for real parser bugs found in development.

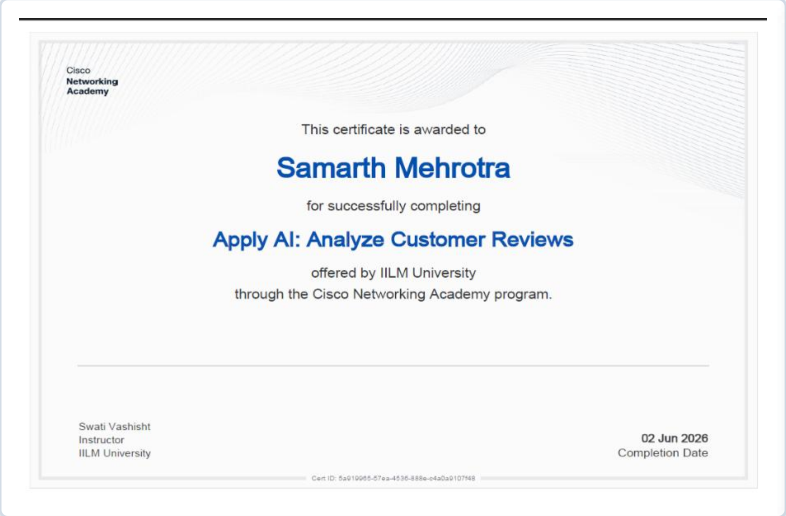
# Certifications Completed

*All offered by IILM University through the Cisco Networking Academy program.*



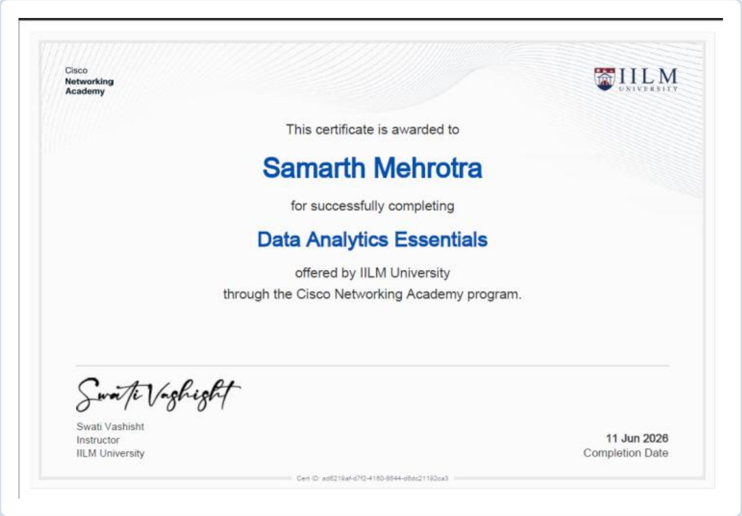

**Introduction to Modern AI**

Completed 02 Jun 2026




**Apply AI: Analyze Customer Reviews**

Completed 02 Jun 2026




**Data Analytics Essentials**

Completed 11 Jun 2026

*The internship's main course is Modern AI — this is an AI-domain submission, so no Packet Tracer (.pkt) file is required.*

# Let the model reason — but verify its evidence automatically, and require a human to approve every action.

*NetSage AI’s most important behaviour is the one thing it refuses to do.*

## Thank You

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